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## Introduction

**Reactive Limitation:** Current ADAS respond only to imminent danger, leaving insufficient human reaction time.

**Proactive Paradigm:** Utilizes Edge-AI to predict vehicle trajectories 3-5 seconds ahead, shifting from crash mitigation to total prevention.

**Vision 2030 Alignment:** Supports the Quality of Life Program by reducing traffic fatalities and advancing Smart City V2X infrastructure.

## Objectives

- **Proactive AI:** Predict hazards 3-5s ahead via LSTM.
- **Edge Processing:** Ensure <150ms latency via Jetson & ROS 2.
- **Context Alerts:** Issue tiered, AI-driven driver warnings.
- **V2X Integration:** Design modular architecture for Smart Cities.
- **Auto-Response:** Trigger instant emergency protocols.

## Methodology

- Data Generation (CARLA Simulator)**  
Extracted 15.3K images for perception and 330K kinematic time-series across 4 critical crash scenarios.
- Vision Perception (YOLOv8n)**  
Trained edge-optimized vehicle detection model, achieving high accuracy with 0.943 mAP@50 and 0.948 Precision.
- Risk Assessment (CNN-LSTM Fusion)**  
Analyzes a 50-frame rolling window of 6 dynamic variables (Speed,  $dv_x$ ,  $dv_y$ ,  $a_x$ ,  $a_y$ , Jerk) to predict collision probability.
- System Integration (ROS 2 Humble)**  
Developed a decentralized node architecture to synchronize perception and prediction modules, initially validated in CARLA simulation.
- Edge Deployment & Hardware Testing**  
Deployed the complete end-to-end ROS 2 system on an NVIDIA Jetson Orin Nano (8GB) to evaluate real-world inference capabilities.

## System Block Diagram

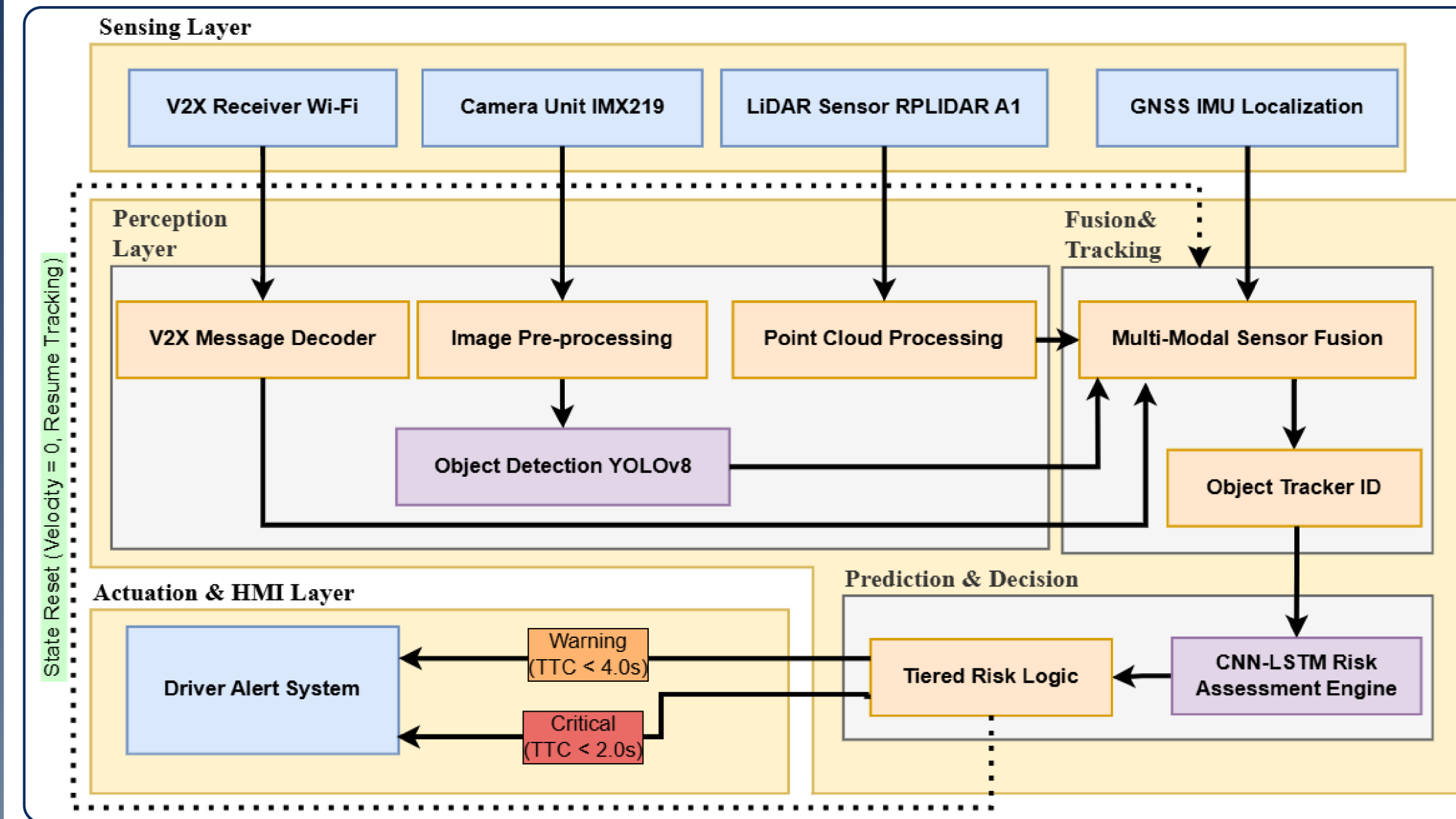


Figure 1: Functional Architecture of Integrated Multi-Modal Perception and Risk Logic

## Results

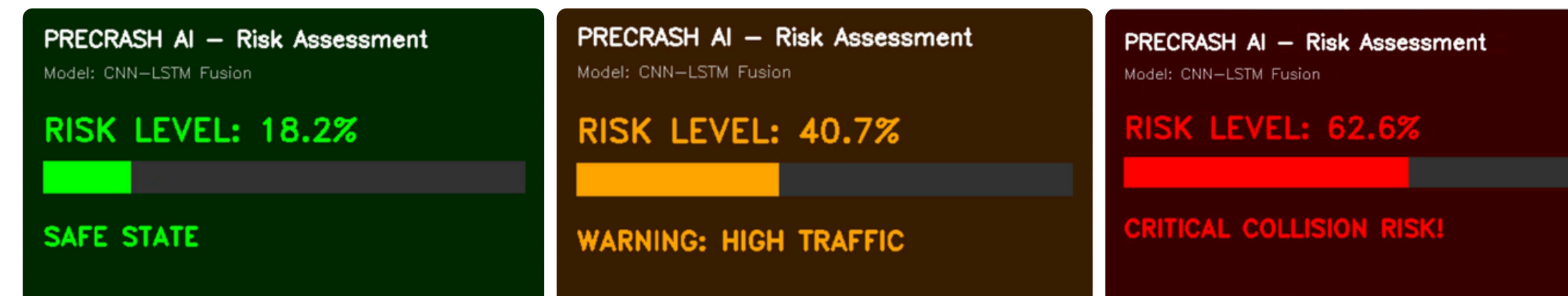


Figure 2: Real-Time Risk Assessment Progression (CNN-LSTM Fusion)



Figure 3: Integrated System: Robot and Interactive Dashboard.

## Performance

### Quantitative Results (YOLOv8n):

- Precision (P): 0.948
- Recall (R): 0.877
- MAP (mAP50): 0.943

### Quantitative Results (LSTM):

- Avg. Disp. Error (ADE): 0.5504 m
- Final Disp. Error (FDE): 0.2865 m
- AUC-ROC: 0.870

### Edge Optimization Highlights

- 22.7x Speedup in overall performance.
- Latency Cut from 312.45 ms to 13.72 ms.
- FPS Boosted from 3.2 to 72.8 (Real-time).

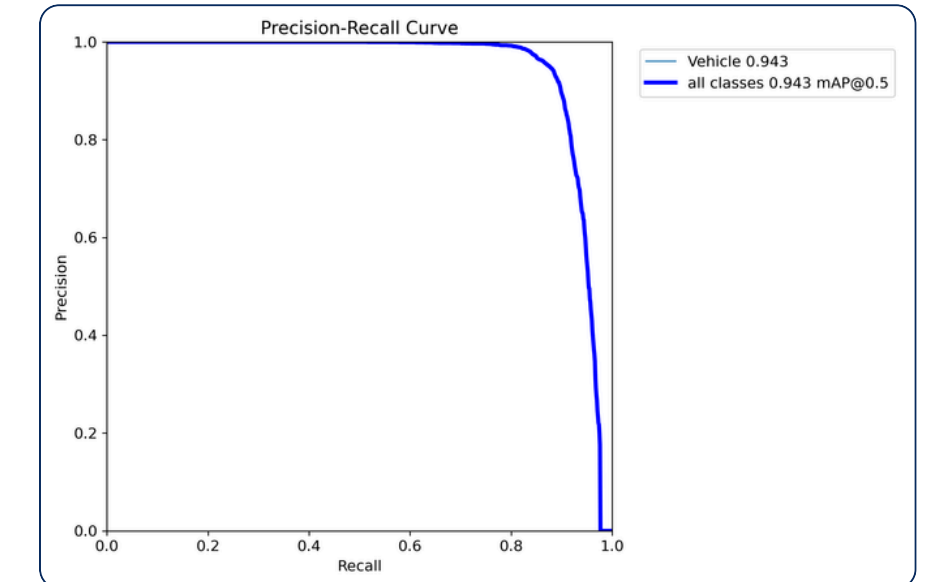


Figure 4: YOLOv8 performance metrics.

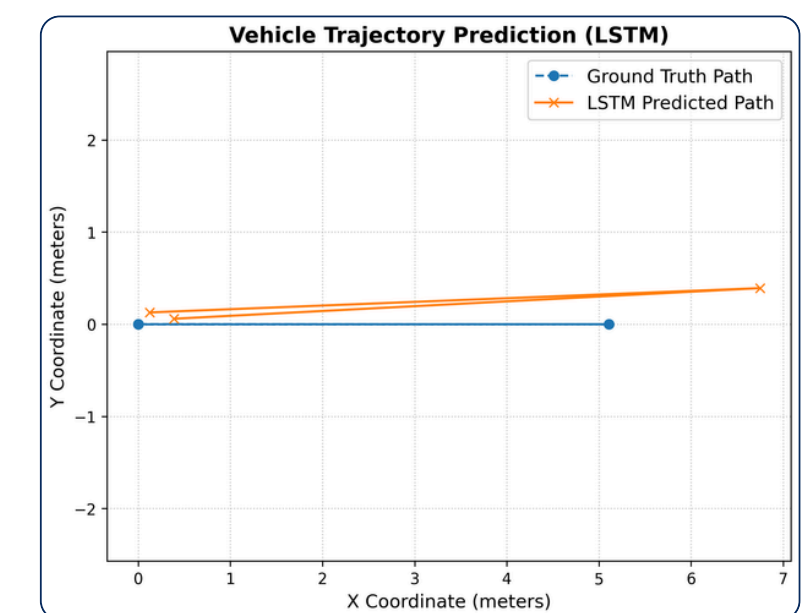


Figure 5: LSTM error metrics.

## Originality & Comparison

| Reference / Study   | Core Approach        | Processing Env.    | Timing / Horizon       |
|---------------------|----------------------|--------------------|------------------------|
| Aleksa et al. [1]   | Statistical Analysis | N/A (Analytical)   | Post-crash             |
| Siswanto et al. [2] | Random Forest        | Cloud Computing    | Post-crash             |
| Malayil et al. [5]  | CNN / ANN            | Gen. Embedded      | At Collision           |
| PRECASH AI (Ours)   | YOLO + LSTM Fusion   | Edge (Jetson Nano) | Proactive (3-5s ahead) |

## Conclusion & Future Work

PRECASH AI provides proactive, real-time safety via an Edge-based YOLOv8/LSTM/ROS 2 system. Future work focuses on intelligent downsampling, and full-scale vehicle testing.

## References

- [1] M. Aleksa et al., "Impact analysis of Advanced Driver Assistance Systems (ADAS) regarding road safety – computing reduction potentials," Eur. Transp. Res. Rev., vol. 16, no. 1, p. 39, Jun. 2024.
- [2] J. Siswanto et al., "Artificial Intelligence in Road Traffic Accident Prediction," Jambura J. Inform., vol. 5, no. 2, pp. 77-90, Nov. 2023.
- [3] M. A et al., "Deep Learning-based Auto Accident Detection and Alert System for Vehicles," in 2023 Int. Conf. Innov. Eng. Technol. (ICIET), Jul. 2023, pp. 1-5.

